SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Initialization and Planning Phase

Date	17 June 2026
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to improve the prediction of the **Human Development Index (HDI)** using Machine Learning, increasing prediction accuracy and supporting better decision-making. It addresses the challenges of manually analyzing multiple development indicators and provides a faster, more reliable, and user-friendly prediction system. Key features include a Machine Learning-based HDI prediction model and real-time prediction through a Flask web application.

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based Human Development Index (HDI) prediction system that provides fast and accurate predictions using key development indicators.
Scope	The project focuses on collecting HDI data, preprocessing it, training a Machine Learning model, and developing a Flask web application for real-time HDI prediction and analysis.
Problem Statement	
Description	Analyzing Human Development Index manually is difficult because it depends on multiple indicators such as Life Expectancy, Education, and Gross National Income (GNI) per capita, making the process time-consuming and complex.
Impact	Solving this problem helps governments, researchers, and policymakers make faster and more accurate development decisions while identifying areas that require improvement.
Proposed Solution	
Approach	Apply Machine Learning algorithms to analyze HDI data and predict the Human Development Index based on Life Expectancy, Education, and GNI through a Flask web application.
Key Features	• Machine Learning-based HDI prediction model. • Real-time prediction of HDI score and category.

	• Simple and user-friendly web interface. • Accurate analysis using development indicators. • Supports informed policy and development planning.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	Processor specifications	Intel Core i3/i5 or above
Memory	RAM specifications	Minimum 4 GB
Storage	Disk space for data, models, and application	Minimum 10 GB Free Storage
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Anaconda, Jupyter Notebook, VS Code / PyCharm
Data		
Data	Source, size, format	HDI Dataset (Kaggle/UNDP), CSV Format